



## Time Series Analysis

### Chapter 4: Seasonality and forecasting: SARIMA, TBATS, Prophet



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## Acronyms used in this material (1/2)

- ACF: Autocorrelation Function
- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model)
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ARIMA: AutoRegressive Integrated Moving Average (model)
- ARMA: AutoRegressive Moving Average (model)
- BIC: Bayesian Information Criterion
- CA: Calendar Adjusted
- DHR: Dynamic Harmonic Regression
- DM: Diebold–Mariano (test)
- ENTSO-E: European Network of Transmission System Operators for Electricity
- ETS: Error, Trend, Seasonality (exponential smoothing state space models)
- EU: European Union
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GDP: Gross Domestic Product
- GW: Gigawatt
- HEGY: Hylleberg–Engle–Granger–Yoo (seasonal unit-root test)
- HICP: Harmonised Index of Consumer Prices
- HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- KPSS: Kwiatkowski–Phillips–Schmidt–Shin (test)
- LB: Ljung–Box (test)
- MA: Moving Average
- MAE: Mean Absolute Error
- MAPE: Mean Absolute Percentage Error
- MASE: Mean Absolute Scaled Error

## Acronyms used in this material (2/2)

- ▣ **MSTL: Multiple Seasonal-Trend decomposition using Loess**
- ▣ **MW: Megawatt**
- ▣ **NSA: Not Seasonally Adjusted**
- ▣ **OCSB: Osborn–Chui–Smith–Birchenhall (seasonal unit-root test)**
- ▣ **PACF: Partial AutoCorrelation Function**
- ▣ **RMSE: Root Mean Squared Error**
- ▣ **SARIMA: Seasonal ARIMA (model)**
- ▣ **SCA: Seasonally and Calendar Adjusted**
- ▣ **SEATS: Signal Extraction in ARIMA Time Series**
- ▣ **STL: Seasonal and Trend decomposition using Loess**
- ▣ **TBATS: Trigonometric seasonality, Box–Cox, ARMA errors, Trend, Seasonal components (model)**
- ▣ **TRAMO-SEATS: Time series Regression with ARIMA noise, Missing values and Outliers; Signal Extraction in ARIMA Time Series**
- ▣ **US: United States**
- ▣ **UTC: Coordinated Universal Time (Temps Universel Coordonné)**
- ▣ **VAT: Value Added Tax**

## Today's question and route

- **Question:** many economic series repeat a pattern every year, week or day; how do we model this pattern, forecast with it, and decide which forecast is better?
  - ▶ three answers: **difference** the pattern away (SARIMA), **model** it with regressors (Fourier terms, TBATS, Prophet), or **remove** it (seasonal adjustment)
- **Route** of the chapter
  - ▶ seasonal differences and seasonal unit roots (HEGY, Canova–Hansen, OCSB)
  - ▶  $\text{SARIMA}(p, d, q)(P, D, Q)_s$ : identification, estimation, diagnostics, forecasts; the airline model and Romanian GDP
  - ▶ seasonal adjustment in practice (X-13ARIMA-SEATS, Eurostat), Fourier terms and calendar effects (Orthodox Easter)
  - ▶ multiple seasonality in electricity load: MSTL, dynamic harmonic regression, TBATS, Prophet
  - ▶ forecast evaluation (time-series cross-validation, Diebold–Mariano test) and forecast combination
- It builds on [Chapter 0](#) (components, Holt–Winters, MASE) and [Chapters 2–3](#) (ARMA, unit roots, ARIMA): the same ideas, at the seasonal lag  $s$
- Seminar 4 comes before this lecture: it gives the formulas; here we derive them and use them on data

## Learning outcomes

- ▣ Recognise deterministic and stochastic seasonality and choose seasonal differencing with HEGY, Canova–Hansen and OCSB
- ▣ Write, identify, estimate, check and forecast a  $SARIMA(p, d, q)(P, D, Q)_s$  model, and expand its polynomials
- ▣ Read seasonally adjusted and unadjusted official series correctly, and explain what X-13ARIMA-SEATS does
- ▣ Model calendar effects (working days, Orthodox Easter) and multiple seasonality with Fourier terms, MSTL, TBATS and Prophet
- ▣ Compare forecasts with time-series cross-validation, MASE and the Diebold–Mariano test, and combine them

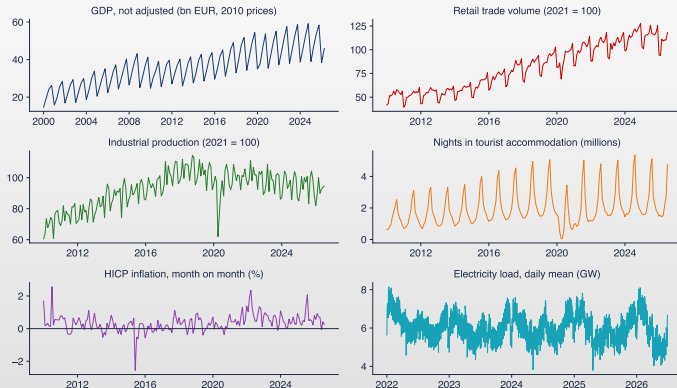
## Reading and tools

- ▣ Textbook: [Huang and Petukhina \(2022\)](#), Ch. 4–5 (ARIMA and nonstationary models)
  - ▶ companion, free online: [Hyndman and Athanasopoulos \(2021\)](#), Ch. 9 (FPP3, [Section 9.9](#)), Ch. 10 (FPP3, [Section 10.5](#)), Ch. 12 (FPP3, [Section 12.1](#)) and Ch. 13
- ▣ The classic: [Box, Jenkins and Reinsel \(2008\)](#), Ch. 9 (seasonal models); seasonal adjustment: [Ladiray and Quenneville \(2001\)](#), [Dagum and Bianconcini \(2016\)](#), [Eurostat \(2024\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch\\_04](#)
  - ▶ statsmodels (SARIMAX, MSTL, ExponentialSmoothing); optional packages tbats and prophet
- ▣ Lecture notebook: [open in Google Colab](#)
- ▣ Video course:  [Applied Time Series Analysis with Python](#)

## Seasonality: definition and sources

- ▣ **Seasonality**: a pattern that repeats with a fixed, known **period**  $s$  (observations per cycle)
  - ▶  $s = 4$  (quarters),  $s = 12$  (months),  $s = 7$  (days in a week),  $s = 24$  and  $s = 168$  (hours in a day and in a week)
  - ▶ unlike a **cycle**, whose length is irregular and unknown ([Chapter 0](#))
- ▣ **Sources**
  - ▶ weather: heating in winter, the beach in August, harvests
  - ▶ the calendar: working days, public holidays, **moving holidays** such as Orthodox Easter
  - ▶ institutions and habits: the school year, tax deadlines, end-of-year bonuses, the working week
- ▣ **Additive** ( $y_t = T_t + S_t + R_t$ ) or **multiplicative** ( $y_t = T_t S_t R_t$ , additive after logs): [Chapter 0](#); here we work with logs when the swings grow with the level

## Seasonality in six Romanian series

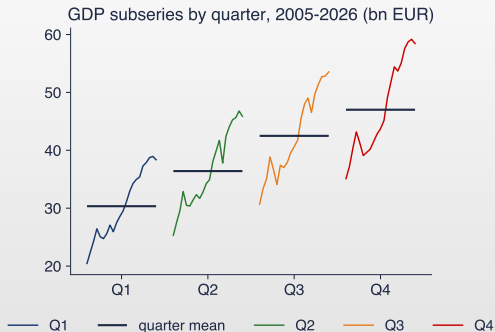
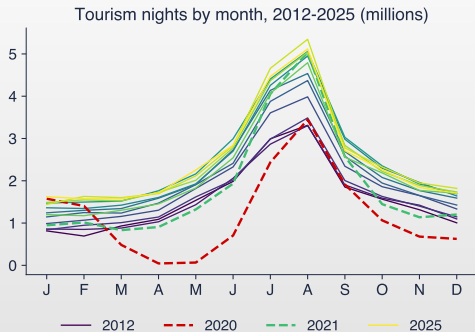


□ Eurostat: GDP (quarterly, not adjusted), retail trade, industrial production, tourism nights, HICP; ENTSO-E: daily electricity load

## Interpreting the six series

- GDP: the mean of Q4 is 57% above the mean of Q1 (2000–2026); the swing grows with the level: multiplicative, take logs
  - ▶ quarter means (bn EUR, 2010 prices): 27.8, 33.6, 39.5, 43.6: construction and agriculture in Q2–Q3
- Tourism: August has 3.7 times the nights of January (2015–2019); the 2020 lockdown breaks the pattern
- HICP: monthly inflation is lower in June ( $-0.37\%$  on average, 2010–2019, fresh food) and higher in October ( $0.49\%$ )
- Electricity: two patterns at once, the week (dense oscillation) and the year (winter peaks): **multiple seasonality**, Section 8

## Seasonal plot and subseries plot



Left: one line per year (seasonal plot); right: each quarter of GDP followed over the years, with its mean (subseries plot)

 TSA\_ch4\_seasonal\_data

## Interpreting the seasonal plots

- Tourism: one shape every year, a summer peak and a winter trough
  - ▶ 2019: 5.06 million nights in August, 1.47 million in January
  - ▶ April 2020: 0.045 million (1.76 million in April 2019): an outlier, not a new season
- GDP subseries: all four quarters rise in parallel
  - ▶ a stable seasonal shape on top of a common trend
  - ▶ if the lines crossed, the seasonal pattern would be changing: stochastic seasonality (next section)



Mamaia, on the Black Sea, September 2013

Photo: Razvan Socol (2013); CC BY-SA 4.0; Wikimedia

[Commons](#)

## Recap: Seasonality in economic data

- Seasonality repeats with a known period  $s$ ; a cycle does not
- Weather, the calendar and institutions create it; moving holidays shift it between months
- Romanian GDP, tourism, inflation and electricity load are all seasonal; electricity has several periods at once
- Seasonal and subseries plots show whether the pattern is stable

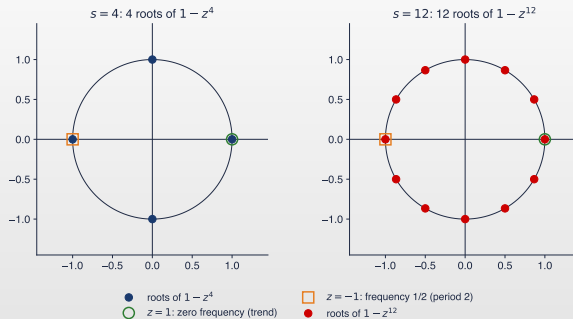
## The seasonal difference

- **Seasonal difference:**  $\Delta_s y_t = (1 - L^s)y_t = y_t - y_{t-s}$ 
  - ▶ with logs:  $100 \Delta_4 \ln Y_t$  = growth over the same quarter of the previous year, in %
- A **fixed** seasonal pattern  $S_t = S_{t-s}$  plus a linear trend:  $y_t = a + bt + S_t + \varepsilon_t$ 
  - ▶  $a$ : the intercept;  $b$ : the slope of the trend;  $S_t$ : the seasonal effect of period  $t$
  - ▶  $\Delta_s y_t = bs + \varepsilon_t - \varepsilon_{t-s}$ :  $\Delta_s$  removes the pattern and turns the trend into a constant
  - ▶ but it creates the non-invertible MA term  $\varepsilon_t - \varepsilon_{t-s}$ : **seasonal over-differencing** (as in [Chapter 3](#))
- $\Delta\Delta_s = (1 - L)(1 - L^s) = 1 - L - L^s + L^{s+1}$ : needed when the trend is stochastic as well
  - ▶ the order of the two differences does not matter: the operators commute

## Deterministic and stochastic seasonality

- **Deterministic:**  $y_t = \sum_{j=1}^s \gamma_j D_{jt} + u_t$ , with  $D_{jt} = 1$  in season  $j$  and  $u_t$  stationary
  - ▶ the seasonal means  $\gamma_j$  never change; the right tool: seasonal dummies or Fourier terms (Section 7)
- **Stochastic:** the **seasonal random walk**  $y_t = y_{t-s} + \varepsilon_t$ 
  - ▶ each season is its own random walk:  $s$  separate walks interleaved
  - ▶  $\text{Var}(y_t)$  grows without bound; the pattern drifts, and “summer can become winter”
  - ▶ the right tool: the seasonal difference  $\Delta_s$
- The choice matters: differencing a deterministic pattern over-differences; modelling a stochastic one with dummies leaves a unit root in the residuals

## The roots of $1 - z^s$ : seasonal unit roots

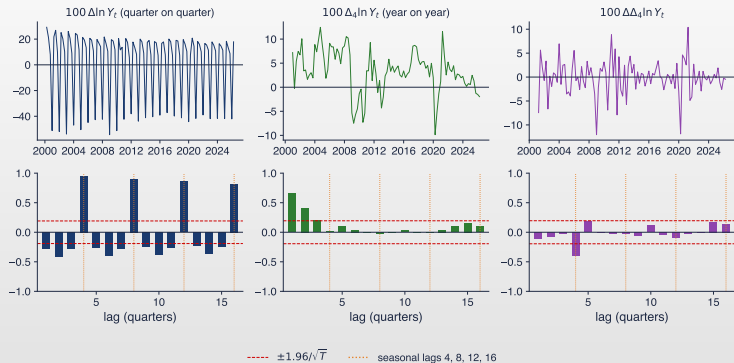


□  $1 - z^4 = (1 - z)(1 + z)(1 + z^2)$ : roots 1,  $-1$ ,  $\pm i$ , all on the unit circle

## Interpreting the seasonal roots

- $\Delta_s$  imposes  $s$  unit roots at once, one for each **frequency**  $k/s$ ,  $k = 0, \dots, s - 1$ 
  - ▶ quarterly:  $z = 1$  (frequency 0, the trend),  $z = -1$  (half a cycle per quarter: period 2 quarters),  $z = \pm i$  (period 4 quarters: the annual cycle)
- A series may have some of these roots and not others: then  $\Delta_4$  over-differences at the frequencies it does not need
- Seasonal unit-root tests ask, frequency by frequency, which roots are present: the HEGY test (next slides)

## Romanian GDP: regular and seasonal differences



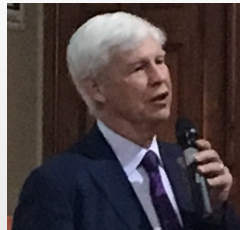
- In of real GDP, not adjusted, 106 quarters to 2026Q2; top: three transformations; bottom: their sample ACF up to 16 quarters

## Interpreting the three transformations

- $\Delta \ln Y$ : dominated by the season: ACF 0.95 at lag 4 and 0.90 at lag 8, decaying very slowly; sd 26.5%
  - ▶ the seasonal analogue of the slow ACF decay of a random walk ([Chapter 3](#))
- $\Delta_4 \ln Y$  (year-on-year growth): sd 4.4%; ACF 0.66 at lag 1, 0.01 at lag 4: an AR-type pattern
- $\Delta\Delta_4 \ln Y$ : ACF  $-0.11$  at lag 1 and  $-0.40$  at lag 4 (band  $\pm 0.20$ )
  - ▶ one negative spike at the seasonal lag: a seasonal MA(1), the signature of the airline model (Section 3)

## The HEGY test

- **Hylleberg, Engle, Granger and Yoo (1990)**: one regression tests every seasonal root of a quarterly series
  - ▶ filters:  $y_{1t} = (1 + L + L^2 + L^3)y_t$ ,  $y_{2t} = -(1 - L + L^2 - L^3)y_t$ ,  
 $y_{3t} = -(1 - L^2)y_t$
- $\Delta_4 y_t = \pi_1 y_{1,t-1} + \pi_2 y_{2,t-1} + \pi_3 y_{3,t-2} + \pi_4 y_{3,t-1} + \text{deterministic terms} + \text{lags of } \Delta_4 y_t + \varepsilon_t$ 
  - ▶  $y_{1t}, y_{2t}, y_{3t}$ : the series filtered so that each keeps one seasonal frequency;  
 $\pi_k$ : the coefficient of each frequency
  - ▶  $H_0$ :  $\pi_1 = 0$  (root 1),  $t$ -test;  $\pi_2 = 0$  (root  $-1$ ),  $t$ -test;  $\pi_3 = \pi_4 = 0$  (roots  $\pm i$ ),  $F$ -test
  - ▶ non-standard distributions, as for Dickey–Fuller: critical values from tables or by simulation
- Rejecting all three: no seasonal differencing needed; rejecting none:  $\Delta_4$  is justified



Robert Engle, a co-author of HEGY, 2017

Photo: Econterms (2017); CC BY-SA 4.0;  
 Wikimedia Commons

## Canova–Hansen and OCSB

- ▣ **Canova–Hansen** (Canova and Hansen (1995)): the null is **stable** (deterministic) seasonality
  - ▶ the seasonal analogue of KPSS (Chapter 3): rejecting means the pattern changes over time
- ▣ **OCSB** (Osborn, Chui, Smith and Birchenhall (1988)): the null is a seasonal unit root; a regression of  $\Delta\Delta_s y_t$  on  $\Delta_s y_{t-1}$  and  $\Delta y_{t-s}$ 
  - ▶ the default seasonal test of `auto_arima` in the Python package `pmdarima` (`nsdiffs`)
- ▣ As in Chapter 3: tests with opposite nulls answer different questions; use them together
  - ▶ and look at the data: seasonal plots, the ACF at lags  $s, 2s, 3s$ , and the size of  $\hat{\Theta}$  after  $\Delta_s$  ( $\hat{\Theta} \approx -1$  signals over-differencing)

## Seasonal unit-root tests on Romanian data

| HEGY, ln GDP, 2000–2019           | statistic | 5% critical value | decision     |
|-----------------------------------|-----------|-------------------|--------------|
| $t(\pi_1)$ , root 1               | −1.97     | −3.39             | not rejected |
| $t(\pi_2)$ , root −1              | −2.19     | −2.77             | not rejected |
| $F(\pi_3, \pi_4)$ , roots $\pm i$ | 6.12      | 6.64              | not rejected |

| Series (logs, 2005–2019) | $D$ by OCSB | $D$ by Canova–Hansen | $d$ by KPSS |
|--------------------------|-------------|----------------------|-------------|
| GDP (quarterly)          | 1           | 0                    | 1           |
| Retail trade             | 0           | 0                    | 1           |
| Industrial production    | 0           | 0                    | 1           |
| Tourism nights           | 0           | 0                    | 1           |
| HICP                     | 0           | 0                    | 2           |

- HEGY: constant, seasonal dummies and trend, 1 lag of  $\Delta_4 y$  (AIC),  $n = 75$ ; critical values from 4 000 simulations of  $\Delta_4 y_t = \varepsilon_t$
- Interpretation: for GDP, HEGY and OCSB support  $\Delta_4$  while Canova–Hansen does not reject a stable pattern; with 20 years of quarters both views fit, so we use  $\Delta_4$  and let  $\hat{\Theta}$  say how stable the pattern is (Section 5)
- Monthly series: both tests choose  $D = 0$ , a stable pattern; HICP needs  $d = 2$  in logs: inflation itself is close to a unit root ([Chapter 3](#))

## Recap: Seasonal differencing and seasonal unit roots

- ▣  $\Delta_s = 1 - L^s$  has  $s$  unit roots on the circle, one per seasonal frequency
- ▣ Deterministic seasonality: dummies or Fourier terms; stochastic seasonality:  $\Delta_s$
- ▣ HEGY and OCSB test for seasonal unit roots; Canova–Hansen tests for a stable pattern
- ▣ Romanian GDP:  $\Delta\Delta_4 \ln Y$  leaves one spike at lag 4, a seasonal MA

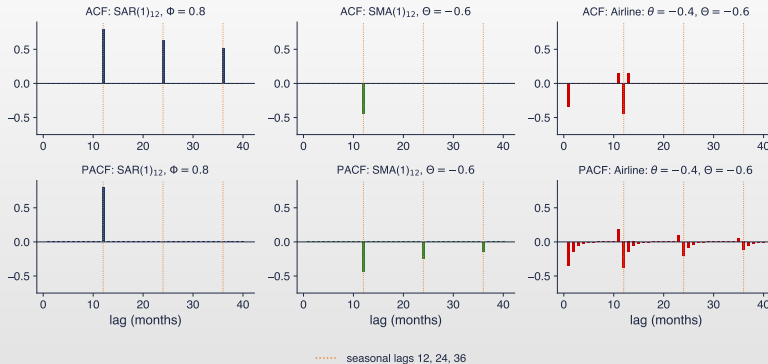
## The SARIMA( $p, d, q$ )( $P, D, Q$ ) $_s$ model

- $\phi(L) \Phi(L^s) (1 - L)^d (1 - L^s)^D y_t = c + \theta(L) \Theta(L^s) \varepsilon_t, \varepsilon_t \sim \text{WN}(0, \sigma^2)$ 
  - ▶ regular polynomials  $\phi(L)$  (order  $p$ ),  $\theta(L)$  (order  $q$ ), as in [Chapter 2](#)
  - ▶ seasonal polynomials in  $L^s$  (orders  $P, Q$ ):  $\Phi(L^s) = 1 - \Phi_1 L^s - \dots - \Phi_P L^{Ps}$ ,  $\Theta(L^s) = 1 + \Theta_1 L^s + \dots + \Theta_Q L^{Qs}$
  - ▶  $d, D$ : numbers of regular and seasonal differences, usually 0 or 1
- Why multiply? [Box, Jenkins and Reinsel \(2008\)](#): in a monthly series, January depends on December (the regular part) and on last January (the seasonal part)
  - ▶ a two-way table: months in the columns, years in the rows; one model along each direction
- Stationarity: the roots of  $\phi(z)\Phi(z^s)$  outside the unit circle; invertibility: the roots of  $\theta(z)\Theta(z^s)$  outside; for a seasonal AR(1) this means  $|\Phi| < 1$

## Worked example: the airline model

- SARIMA(0, 1, 1)(0, 1, 1)<sub>12</sub> with  $\theta = -0.4$ ,  $\Theta = -0.6$ :  $(1 - L)(1 - L^{12})y_t = (1 - 0.4L)(1 - 0.6L^{12})\varepsilon_t$ 
  - ▶ left:  $y_t - y_{t-1} - y_{t-12} + y_{t-13}$ ; right:  $\varepsilon_t - 0.4\varepsilon_{t-1} - 0.6\varepsilon_{t-12} + 0.24\varepsilon_{t-13}$
  - ▶ the coefficient of  $\varepsilon_{t-13}$ ,  $\theta\Theta = 0.24$ , is not a new parameter: two parameters plus  $\sigma^2$
- $w_t = \Delta\Delta_{12}y_t$  is an MA(13) with only four nonzero weights; its autocovariances ([Chapter 2](#)):
  - ▶  $\gamma(0) = \sigma^2(1 + \theta^2)(1 + \Theta^2) = 1.5776\sigma^2$
  - ▶  $\rho_1 = \theta/(1 + \theta^2) = -0.345$ ;  $\rho_{12} = \Theta/(1 + \Theta^2) = -0.441$ ;  $\rho_{11} = \rho_{13} = \rho_1\rho_{12} = 0.152$
  - ▶ all other autocorrelations are zero

## Theoretical ACF and PACF of three seasonal models



□ Monthly,  $s = 12$ ; the airline panel is the stationary part  $w_t = \Delta\Delta_{12}y_t$

## Interpreting the seasonal patterns

- Seasonal AR(1): the ACF decays geometrically at lags 12, 24, 36 (0.8, 0.64, 0.512); the PACF has one spike at lag 12
  - ▶ the AR rule of [Chapter 2](#), read only at multiples of  $s$
- Seasonal MA(1): the ACF cuts off after lag 12; the PACF decays at 12, 24, 36 ( $-0.242$  at lag 24)
- Airline: spikes at 1 ( $-0.345$ ) and 12 ( $-0.441$ ), with “satellites” at 11 and 13 (0.152)
  - ▶ the satellites are the product  $\rho_1 \rho_{12}$ : the fingerprint of the multiplicative structure

## Identification of SARIMA models

| After differencing | ACF                                | PACF                               |
|--------------------|------------------------------------|------------------------------------|
| seasonal AR( $P$ ) | decays at $s, 2s, \dots$           | cuts off after lag $Ps$            |
| seasonal MA( $Q$ ) | cuts off after lag $Qs$            | decays at $s, 2s, \dots$           |
| regular part       | lags $1, 2, \dots$ as in Chapter 2 | lags $1, 2, \dots$ as in Chapter 2 |
| product of both    | satellites at $ks \pm j$           | satellites at $ks \pm j$           |

### □ Steps

- ▶ 1. Stabilise the variance (logs or Box–Cox, Box and Cox (1964)); 2. choose  $D$  (seasonal plot, HEGY, OCSB), then  $d$  (Chapter 3)
- ▶ 3. Read  $P, Q$  at the seasonal lags and  $p, q$  at the first lags; 4. keep the orders small:  $P, Q \leq 1$  almost always suffice

## Estimation and model choice

- ▣ **Maximum likelihood** of the differenced series (state-space form, `statsmodels SARIMAX`)
  - ▶  $s + 1$  observations are lost to  $\Delta\Delta_s$ ; the constant is dropped when  $d + D \geq 2$  (it would mean a quadratic trend)
- ▣ **Information criteria:**  $AICc = AIC + \frac{2k(k+1)}{n-k-1}$  and BIC, compared **only** across models with the same  $d$  and  $D$ 
  - ▶  $k$ : the number of estimated parameters;  $n$ : the number of observations after differencing
  - ▶ differencing changes the data on which the likelihood is computed
- ▣ **Automatic SARIMA:** a stepwise search over  $(p, q, P, Q)$  by AICc after choosing  $d$  and  $D$  by tests ([Hyndman and Khandakar \(2008\)](#))
  - ▶ in Python: `pmdarima.auto_arima(y, m=12)`; always check the residuals and compare with a benchmark
- ▣ **Diagnostics:** Ljung–Box with lags beyond  $s$  (e.g.  $m = 2s$ ) and  $m - p - q - P - Q$  degrees of freedom ([Ljung and Box \(1978\)](#))

## Forecasting with SARIMA

- As for ARIMA (Chapter 3): write the model as an equation for  $y_t$ ; replace future shocks by 0, future values by their forecasts, past shocks by residuals
  - ▶ airline:  $\hat{y}_{T+h} = \hat{y}_{T+h-1} + \hat{y}_{T+h-s} - \hat{y}_{T+h-s-1} + (\text{MA terms while } h \leq s + 1)$
- **Eventual forecast function** of the airline model: a fixed seasonal pattern on a straight line
  - ▶ the pattern is an exponentially weighted average of past patterns, with weight  $1 + \Theta$  on the most recent year
  - ▶  $\Theta \rightarrow -1$ : a fixed (deterministic) pattern;  $\Theta = 0$ : last year's pattern repeated (seasonal naive)
- Intervals widen with  $h$  (two unit roots); for  $\ln y$ ,  $\exp(\hat{y}_{T+h})$  is the forecast **median**
  - ▶ the mean is  $\exp(\hat{y}_{T+h} + \sigma_h^2/2)$ , slightly higher;  $\sigma_h^2$ : the forecast error variance at horizon  $h$

## Worked example: forecasting a seasonal MA

- Quarterly SARIMA(0,0,0)(0,1,1)<sub>4</sub>:  $y_t = y_{t-4} + \varepsilon_t + \Theta \varepsilon_{t-4}$ ,  $\Theta = -0.6$ 
  - ▶ last year:  $y_{T-3}, \dots, y_T = 100, 120, 130, 140$ ; residuals  $\hat{\varepsilon}_{T-3}, \dots, \hat{\varepsilon}_T = 2, -1, 0, 3$
- $\hat{y}_{T+1} = y_{T-3} + \Theta \hat{\varepsilon}_{T-3} = 100 - 0.6 \cdot 2 = 98.8$ 
  - ▶  $\hat{y}_{T+2} = 120.6$ ,  $\hat{y}_{T+3} = 130.0$ ,  $\hat{y}_{T+4} = 138.2$
- $\hat{y}_{T+5} = \hat{y}_{T+1} = 98.8$ : after one year the MA term is gone and the forecast pattern repeats
  - ▶ a large positive surprise last season ( $\hat{\varepsilon}_T = 3$ ) lowers the forecast for that season by  $0.6 \cdot 3$ : partial correction

## Recap: SARIMA models

- ▣ SARIMA multiplies regular and seasonal polynomials:  $\phi(L)\Phi(L^s)\Delta^d\Delta_s^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$
- ▣ Identification: the [Chapter 2](#) rules at lags  $s, 2s, \dots$ , plus satellites at  $ks \pm j$
- ▣ Compare AICc only for the same  $d$  and  $D$ ; check Ljung–Box beyond lag  $s$
- ▣ The airline forecast: a seasonal pattern on a line, updated with weight  $1 + \Theta$

## Box and Jenkins and the airline passengers

- **Box, Jenkins and Reinsel (2008)** (first edition 1970) modelled the monthly totals of international airline passengers, 1949–1960 (144 months)
  - ▶ growth plus a summer peak whose size grows with the level
  - ▶ their model,  $SARIMA(0, 1, 1)(0, 1, 1)_{12}$  for  $\ln y_t$ , has carried the name “airline model” ever since
- Why it lasted
  - ▶ two parameters fit many seasonal series: trade, production, retail sales, energy
  - ▶ it is the reference model of the regARIMA step of seasonal-adjustment software (Section 6)

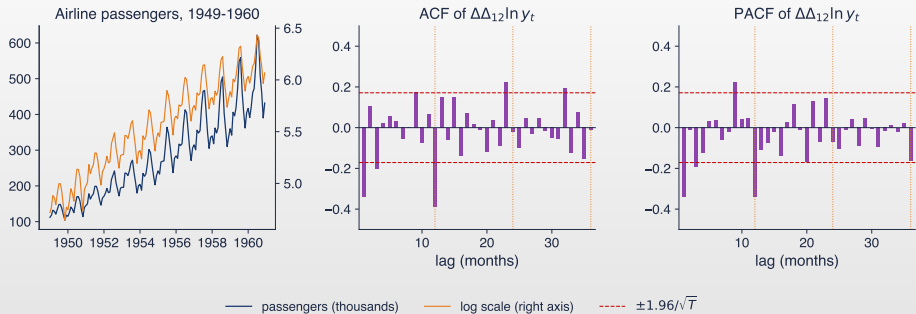


The first Boeing 707s for Pan American, Seattle, 1958

Photo: unknown author (1958); public domain; [Wikimedia](#)

[Commons](#)

## Identification of the airline model

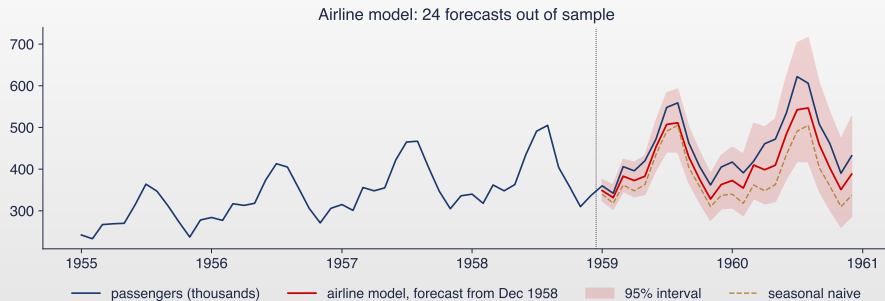


□ Left: passengers and their log; middle and right: ACF and PACF of  $w_t = \Delta\Delta_{12}\ln y_t$ , 131 observations

## Interpreting the airline correlograms

- The log makes the summer swing constant: the multiplicative pattern becomes additive
- ACF of  $w_t$ :  $-0.34$  at lag 1 and  $-0.39$  at lag 12, outside the band  $\pm 0.17$ 
  - ▶ a regular MA(1) and a seasonal MA(1); lag 3 ( $-0.20$ ) is borderline
- The PACF decays at lags 1, 2, 3 and 12, 24: consistent with MA terms, not AR terms
- Conclusion: SARIMA(0, 1, 1)(0, 1, 1)<sub>12</sub>, exactly the model of Box and Jenkins

## Airline model: forecasts out of sample



Estimated on 1949–1958, forecasts for the 24 months of 1959–1960, back-transformed from  $\ln y$

## Interpreting the airline forecasts

- 1949–1958:  $\hat{\theta} = -0.342$  (0.087),  $\hat{\Theta} = -0.541$  (0.105); full sample:  $\hat{\theta} = -0.402$  (0.073),  $\hat{\Theta} = -0.557$  (0.096)
  - ▶ residual sd 3.7% per month; Ljung–Box  $Q^*(24)$  on 22 degrees of freedom,  $p = 0.37$
- MAPE over 1959–1960: 8.5% for the airline model, 15.5% for the seasonal naive forecast (which ignores growth)
- December 1960: forecast 388, interval [286, 526], actual 432: inside, but the model under-predicts the 1959–1960 growth
  - ▶ the interval width grows from 51 (first month) to 239 (24th month): two unit roots

## SARIMA models for Romanian GDP

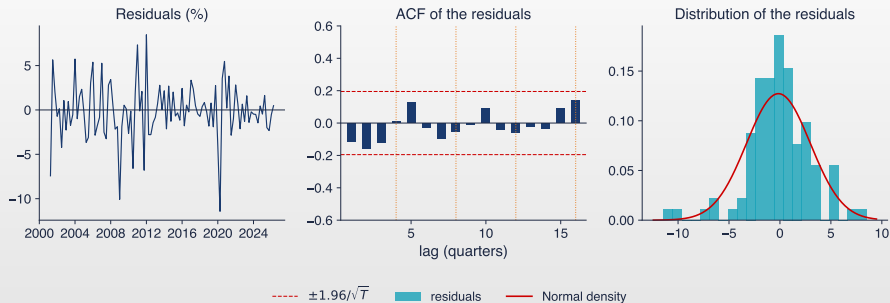
| Model                           | AICc  | BIC   | LB(8), p | $\hat{\sigma}$ (%) |
|---------------------------------|-------|-------|----------|--------------------|
| (0, 1, 1)(0, 1, 1) <sub>4</sub> | 520.1 | 527.6 | 0.157    | 3.05               |
| (0, 1, 0)(0, 1, 1) <sub>4</sub> | 519.7 | 524.8 | 0.286    | 3.08               |
| (1, 1, 0)(0, 1, 1) <sub>4</sub> | 520.7 | 528.3 | 0.156    | 3.06               |
| (1, 1, 1)(0, 1, 1) <sub>4</sub> | 518.3 | 528.4 | 0.467    | 2.98               |
| (0, 1, 2)(0, 1, 1) <sub>4</sub> | 518.8 | 528.8 | 0.575    | 3.00               |
| (0, 1, 1)(1, 1, 0) <sub>4</sub> | 529.9 | 537.5 | 0.003    | 3.22               |
| (0, 1, 1)(1, 1, 1) <sub>4</sub> | 522.2 | 532.2 | 0.118    | 3.05               |
| (0, 1, 1)(0, 1, 0) <sub>4</sub> | 545.1 | 550.2 | 0.005    | 3.52               |

- $y_t = 100 \ln Y_t$ , real GDP not adjusted, 2000Q1–2026Q2,  $T = 106$ ; maximum likelihood; all with  $d = D = 1$ , so the criteria are comparable
- Ljung–Box on 8 lags with  $8 - p - q - P - Q$  degrees of freedom

## Interpreting the model table

- BIC chooses  $(0, 1, 0)(0, 1, 1)_4$ : only the seasonal MA term; AICc chooses  $(1, 1, 1)(0, 1, 1)_4$ 
  - ▶ the AICc model has  $\hat{\phi} = 0.78$  and  $\hat{\theta} = -0.94$ : almost a common factor ([Chapter 2](#)), little gain
  - ▶ the airline model:  $\hat{\theta} = -0.171$  (0.109), not significant;  $\hat{\Theta} = -0.585$
- A seasonal AR instead of the seasonal MA, or no seasonal term at all: much worse criteria and correlated residuals
- We keep the BIC model: one parameter,  $\hat{\Theta} = -0.577$  (0.075),  $\hat{\sigma} = 3.08\%$

## Diagnostics of the GDP model

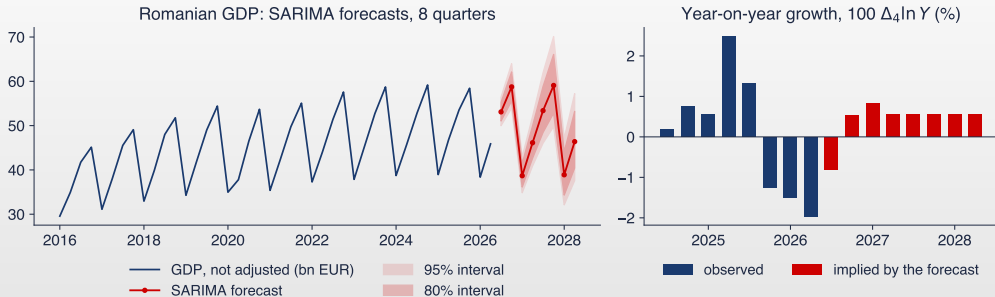


□ SARIMA(0, 1, 0)(0, 1, 1)<sub>4</sub> for 100 ln Y<sub>t</sub>: residuals, their ACF and their distribution

## Interpreting the GDP diagnostics

- No autocorrelation left:  $Q^*(8) = 8.55$  on 7 degrees of freedom,  $p = 0.286$ ;  $Q^*(12)$ :  $p = 0.526$ 
  - ▶ no spike at the seasonal lags 4, 8, 12, 16: the seasonal structure is captured
- Not Normal: Jarque–Bera 22.0 ( $p < 0.001$ ); the largest residual is  $-11.4\%$  in 2020Q2 ( $-3.6$  standard deviations): the lockdown
  - ▶ without 2020, Jarque–Bera falls to 13.4: an additive-outlier dummy would be the next step

## Forecasts of Romanian GDP



□ SARIMA(0, 1, 0)(0, 1, 1)<sub>4</sub>, 8 quarters from 2026Q3 to 2028Q2, 80% and 95% intervals; right: implied year-on-year growth

 TSA\_ch4\_gdp\_case

## Interpreting the GDP forecasts

- Last value: 45.9 bn EUR in 2026Q2; forecasts 53.1, 58.7, 38.7, 46.1 bn EUR for the next four quarters
  - ▶ 95% interval for the first quarter [50.0, 56.4], for the eighth [37.6, 57.2]
- Implied year-on-year growth:  $-0.81\%$ ,  $0.53\%$ ,  $0.83\%$ ,  $0.55\%$ , then a constant  $0.55\%$ 
  - ▶ the model extrapolates the recent weak growth; it knows nothing about fiscal policy or EU funds
- The seasonal pattern of the forecast is a weighted average of past years, with weight  $1 + \hat{\Theta} = 0.42$  on the newest year

## Recap: Airline passengers and Romanian GDP

- ▣ Logs, then  $\Delta\Delta_s$ , then MA terms at lags 1 and  $s$ : the airline recipe works for both series
- ▣ For GDP, BIC drops the regular MA term; AICc picks a near-redundant ARMA(1,1)
- ▣ Residuals are clean except for the 2020 outlier
- ▣ Forecasts repeat a smoothed seasonal pattern on a trend line, with widening intervals

## Why official statistics adjust for seasonality

- ▣ Users want to know whether the economy **grew this quarter**, not whether Q3 is bigger than Q2
  - ▶ 2026Q1: unadjusted GDP changed by  $-42.1\%$  (log points) from the previous quarter; the adjusted series by  $-0.06\%$
  - ▶ the first number is pure season; only the second says something about the business cycle
- ▣ **Seasonal adjustment**: estimate the seasonal and calendar components and remove them:  $y_t^{SA} = y_t - \hat{S}_t - \hat{C}_t$  (or  $y_t / (\hat{S}_t \hat{C}_t)$ )
  - ▶ the classical decomposition and STL of [Chapter 0](#) are simple versions of this
- ▣ For forecasting with SARIMA we use the **unadjusted** series: adjustment filters change the autocorrelations and are revised when new data arrive

## X-11 and X-13ARIMA-SEATS

- **X-11** (US Census Bureau, 1960s; [Ladiray and Quenneville \(2001\)](#)): iterated moving averages
  - ▶ a centred  $2 \times 12$  moving average for the trend, then  $3 \times 3$  and  $3 \times 5$  moving averages of each month across years for the seasonal factors
- **X-12-ARIMA** ([Findley et al. \(1998\)](#)) and **X-13ARIMA-SEATS** add a **regARIMA** pre-adjustment:
  - ▶ regression with SARIMA errors (often the airline model) for outliers, working days, Easter
  - ▶ ARIMA forecasts extend the series, so the symmetric filters also work at the end: smaller revisions
  - ▶ then X-11 filters or SEATS (model-based decomposition of the ARIMA model)
- EU: Eurostat and the national institutes follow [Eurostat \(2024\)](#), with the JDemetra+ software (X-13 and TRAMO-SEATS)



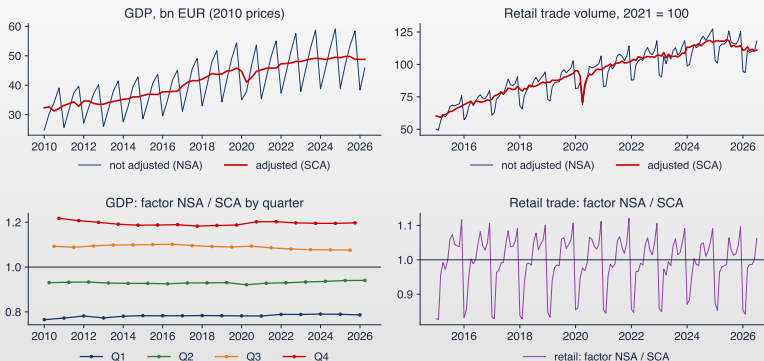
US Census Bureau, Suitland, Maryland:  
the home of X-11 and X-13

Photo: [United States Census Bureau \(2007\)](#);  
public domain; [Wikimedia Commons](#)

## Eurostat series: NSA, CA, SA, SCA

- Each series comes in several versions (dimension `s_adj`):
  - ▶ **NSA**: not seasonally adjusted (the raw data)
  - ▶ **CA**: calendar adjusted only (working days, holidays)
  - ▶ **SA**: seasonally adjusted; **SCA**: seasonally and calendar adjusted
- Which one for which question
  - ▶ quarter-on-quarter or month-on-month growth: SCA
  - ▶ year-on-year growth: NSA or CA (the seasonal pattern cancels, the calendar does not)
  - ▶ SARIMA modelling and forecasting: NSA, with calendar regressors
- The annual totals of NSA and SCA are close (2024: 196.4 and 196.4 bn EUR): adjustment moves output between quarters, it does not create it

## Unadjusted and adjusted series



- Eurostat NSA and SCA: GDP (top left) and retail trade (top right); bottom: the implied seasonal and calendar factors NSA/SCA

## Interpreting the adjusted series

- GDP factors 2015–2019: Q1 0.78, Q2 0.93, Q3 1.10, Q4 1.19: Q1 output is about a fifth below an average quarter
  - ▶ the factors move slowly over the years: the stochastic seasonality that  $\hat{\Theta}$  measures
- Retail trade: December factor 1.10, January 0.84 (2016–2019): Christmas shopping, then the January low
- The retail factor also jumps from month to month because of working days and Easter: the calendar part of SCA (Section 7)
- The adjusted GDP shows the 2020 fall and the 2025–2026 stagnation that the raw series hides

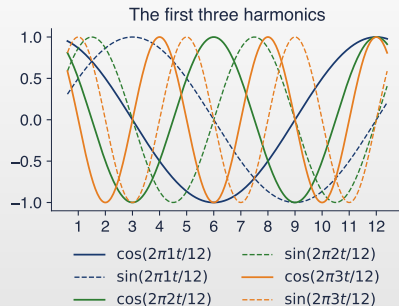
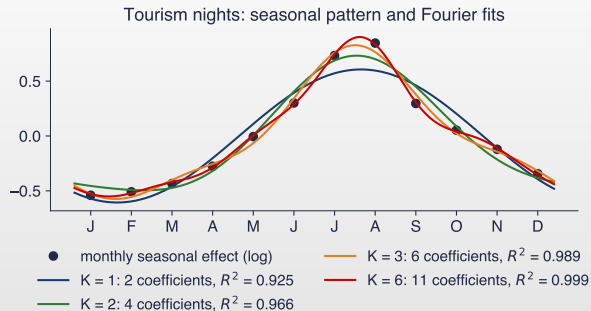
## Recap: Seasonal adjustment

- Seasonal adjustment removes  $\hat{S}_t$  and  $\hat{C}_t$  to show the trend and the cycle
- X-13ARIMA-SEATS: regARIMA pre-adjustment (outliers, calendar, forecasts), then X-11 or SEATS; in the EU, JDemetra+
- Quarter-on-quarter growth: SCA; year-on-year growth: NSA or CA; SARIMA forecasting: NSA

## Seasonal dummies and Fourier terms

- **Dummies:**  $y_t = \beta_0 + \sum_{j=2}^s \gamma_j D_{jt} + u_t$ :  $s - 1$  coefficients, any shape
  - ▶  $\beta_0$ : the level of season 1;  $\gamma_j$ : the difference between season  $j$  and season 1
  - ▶ too many for  $s = 52$  weeks,  $s = 168$  hours or  $s = 365$  days; impossible for  $s = 365.25$
- **Fourier terms** for a period  $m$ :  $\sum_{k=1}^K \left[ \alpha_k \sin \frac{2\pi kt}{m} + \beta_k \cos \frac{2\pi kt}{m} \right]$ 
  - ▶  $\alpha_k, \beta_k$ : the coefficients of the  $k$ -th harmonic, a wave with  $k$  cycles per period;  $K$ : the number of sine–cosine pairs
  - ▶  $2K$  coefficients;  $K = m/2$  reproduces the dummies exactly (for even  $m$ ); a small  $K$  gives a smooth pattern
  - ▶  $m$  may be non-integer, and several periods can be combined: e.g.  $K = 5$  for  $m = 24$  and  $K = 4$  for  $m = 168$  give 18 regressors instead of  $23 + 167$
- $K$  is chosen by AICc or by cross-validation; the same terms are inside TBATS and Prophet (Section 9)

## Fourier approximation of a seasonal pattern



□ Monthly seasonal effect of  $\ln(\text{tourism nights})$ , 2012–2019, after removing a 13-month centred mean; fits with  $K = 1, 2, 3, 6$

 TSA\_ch4\_calendar\_fourier

## Interpreting the Fourier fits

- $K = 1$  (two coefficients) already gives  $R^2 = 0.925$ : the summer peak is close to a single wave
  - ▶  $K = 2$ : 0.966;  $K = 3$ : 0.989 with six coefficients instead of eleven
- $K = 6$  (11 coefficients) interpolates the 12 monthly points: the same as 11 dummies,  $R^2 = 0.999$
- Higher harmonics add sharper detail: the August peak needs  $K \geq 3$

## Calendar effects and Orthodox Easter

- **Working days:** a month with more Monday–Friday non-holiday days produces and sells more
  - ▶ known in advance: a regressor that can be used in forecasts
- **Moving holidays:** Orthodox Easter falls between 4 April and 8 May
  - ▶ 5 May 2024, 20 April 2025, 12 April 2026, 2 May 2027: shopping shifts between March, April and May
  - ▶ in Python: `dateutil.easter.easter(y, EASTER_ORTHODOX)`; Pentecost is 49 days later
- **Easter regressor** (X-13 `easter[w]`):  $E_t$  = share of the  $w$  days before Easter Sunday that fall in month  $t$ 
  - ▶  $w = 10$ : Easter on 20 April gives  $E_{\text{April}} = 1$ ; on 5 May,  $E_{\text{April}} = 0.6$  and  $E_{\text{May}} = 0.4$



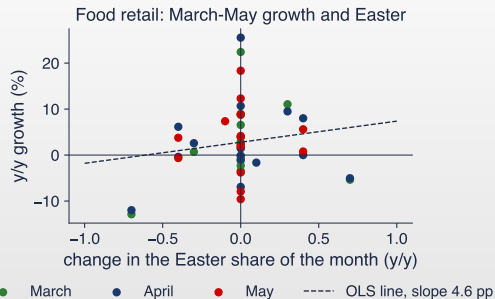
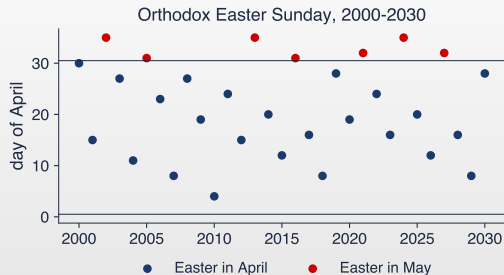
Painted Easter eggs from Romania

Photo: [jackmac34 \(Pixabay\)](#); CC0;  
[Wikimedia Commons](#)

## Regression with SARIMA errors

- $100 \ln y_t = \beta_E E_t + \beta_W WD_t + \text{outlier dummies} + u_t$ , with  $u_t \sim \text{SARIMA}(0, 1, 1)(0, 1, 1)_{12}$ 
  - ▶  $E_t$ : the Easter regressor;  $WD_t$ : the number of working days in month  $t$ ;  $\beta_E, \beta_W$ : their effects, in %
  - ▶ the “regARIMA” model of X-13; estimated in one step by maximum likelihood (SARIMAX with exog)
  - ▶ ordinary least squares with SARIMA errors ignored would give wrong standard errors ([Chapter 3](#), spurious regression)
- Data: Eurostat retail trade volume, Romania, not adjusted, January 2010–July 2026: food (G47\_FOOD) and all retail (G47)
  - ▶ dummies for April and May 2020 (lockdown)

## Easter and Romanian retail trade



- Left: Orthodox Easter Sunday as a day of April (above 30: May); right: year-on-year growth of food retail in March-May against the change in the Easter share (2020–2021 left out)

## Interpreting the Easter effect

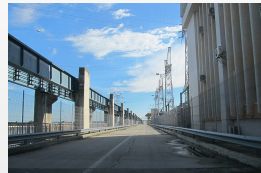
- Food retail:  $\hat{\beta}_E = 5.9\%$  (se 2.3,  $t = 2.6$ ): a month that holds the 10 days before Easter sells about 5.9% more food
  - ▶ one working day more: +0.59% (se 0.17); April 2020: -15%
- All retail: Easter 0.8% (se 1.9), not significant; working days +1.02% (se 0.15)
  - ▶ the Easter effect is in food, the working-day effect in everything
- The raw scatter is noisy (slope 4.6 percentage points): VAT changes and prices also move year-on-year growth; the regression separates the effects
- Easter fell in May in 2002, 2005, 2013, 2016, 2021, 2024: a forecaster who ignores this mistakes April and May

## Recap: Fourier terms and calendar effects

- ▣ Fourier terms describe a seasonal pattern with  $2K$  coefficients, for any period, even non-integer
- ▣ Working days and moving holidays are known in advance: model them as regressors
- ▣ Romanian food retail rises about 5.9% in the Easter month; total retail follows working days
- ▣ Regression with SARIMA errors: estimate regressors and dynamics together

## Electricity load in Romania

- **Load:** the electricity consumed in the system, in MW (hourly averages); it must be matched by generation at every moment
  - ▶ forecasts are needed for every hour of the next day: day-ahead markets, reserves, grid planning
- **Data:** ENTSO-E (the European network of transmission system operators), “monthly hourly load values”, Romania
  - ▶ 39 406 hours, 2022–2026, Romanian winter time (UTC+2) to avoid the clock changes
- Load forecasting has its own competitions: GEFCom2014 ([Hong et al. \(2016\)](#))

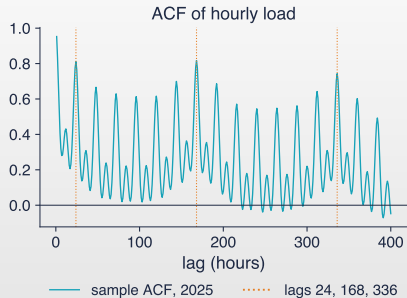
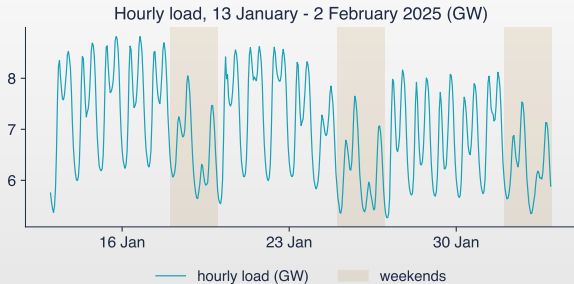


Iron Gates II hydropower plant on the Danube

Photo: [Nenea hartia \(2016\)](#); CC BY-SA 4.0;

[Wikimedia Commons](#)

## Hourly load: the day and the week

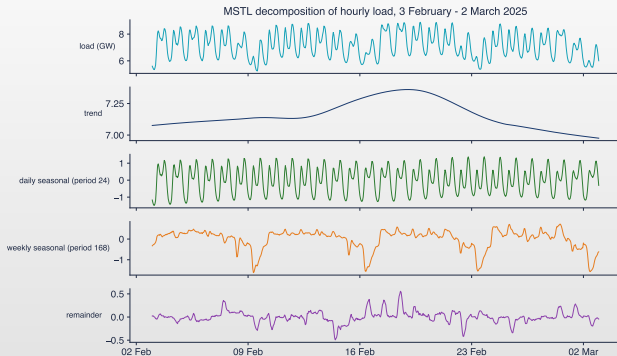


□ Left: three weeks of January 2025; right: sample ACF of all hours of 2025 up to lag 400

## Interpreting the hourly load

- Daily cycle: lowest around 3:00 (5.3 GW on average in 2025), highest around 19:00 (7.3 GW)
  - ▶ ACF 0.81 at lag 24, 0.43 at lag 12
- Weekly cycle: weekends are lower; ACF 0.82 at lag 168 (one week), above the lag-24 value
  - ▶ the same hour one week ago is the best single predictor: Monday is like last Monday, not like Sunday
- And an annual cycle on top (heating and air conditioning): three periods, 24, 168 and about 8766 hours
- SARIMA is not enough: it has one integer period  $s$ , while here  $s = 24, 168$  and 8766 act at once
  - ▶ a polynomial in  $L^{168}$  is slow and unstable, 365.25 is not an integer, and holidays break the weekly pattern
  - ▶ answers: MSTL, the dynamic harmonic regression (DHR), TBATS, Prophet

## MSTL: one seasonal component per period

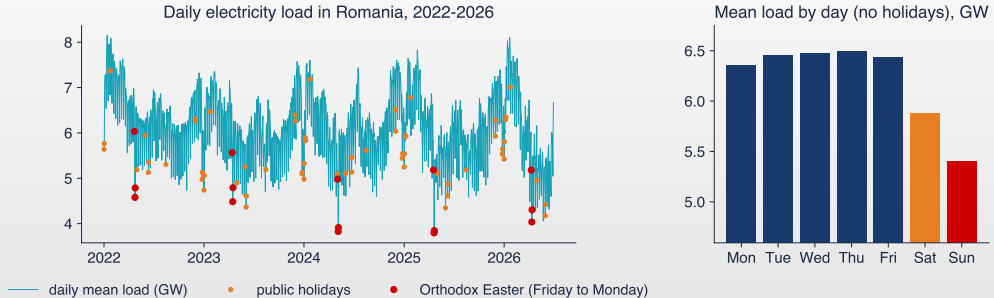


- ❑ MSTL (Bandara, Hyndman and Bergmeir (2025)) applies STL (Chapter 0) repeatedly, once for each period (24 and 168 hours), until the components settle

## Interpreting the MSTL decomposition

- ▣ Of the variance around the trend: 70% daily component, 29% weekly component, 2% remainder
  - ▶ range of the daily component in the first week: 2.8 GW; of the weekly one: 2.1 GW
- ▣ The weekly component is flat Monday to Friday and drops on Saturday and more on Sunday
- ▣ The remainder is small but not white noise: cold spells and holidays; a forecasting model adds dynamics and regressors

## Daily load and public holidays



- Daily mean load, 2022–2026, 1 642 days; Romanian public holidays marked; right: mean by day of the week, holidays excluded

## Interpreting the daily load

- Week: Saturday 9% and Sunday 17% below Wednesday
  - ▶ Year: January 25% above May; a smaller summer peak (air conditioning)
- Holidays: the lowest day of the sample is Easter Sunday, 20 April 2025 (3.79 GW, mean 6.18 GW)
  - ▶ the four Easter days (Friday to Monday) average 4.62 GW; Christmas to New Year is also low
- The level falls over 2022–2026; possible causes: the price shock of 2022 and rooftop solar panels that cover part of the demand behind the meter

## A dynamic harmonic regression for daily load

- $y_t = \sum_{k=1}^4 \left[ \alpha_k \sin \frac{2\pi kt}{365.25} + \beta_k \cos \frac{2\pi kt}{365.25} \right] + \sum_j \delta_j D_{jt}^{\text{day}} + \lambda_1 H_t + \lambda_2 E_t + \lambda_3 X_t + u_t$ 
  - ▶  $\alpha_k, \beta_k$ : annual Fourier coefficients;  $\delta_j$ : effect of weekday  $j$  against Sunday;  $\lambda_1, \lambda_2, \lambda_3$ : effects of the special days, in GW
  - ▶ annual Fourier terms; weekday dummies (Sunday is the base);  $H$  public holiday,  $E$  Easter (Friday to Monday),  $X$  24 December – 2 January
  - ▶  $u_t \sim \text{ARIMA}(1,1,1)$ , chosen by AIC: **dynamic harmonic regression** (DHR, [FPP3, Section 10.5](#))
- Estimates, data to the first forecast origin (30 June 2025), in GW:
  - ▶ Wednesday +1.08, Monday +0.88, Saturday +0.50 relative to Sunday
  - ▶ public holiday −0.38 (se 0.02), Easter −0.58 more, Christmas–New Year −0.61
  - ▶ errors: AR 0.80, MA −0.98,  $\hat{\sigma} = 186$  MW; Ljung–Box(14)  $p < 0.001$ : some dynamics left

## Recap: Multiple seasonality

- Hourly load has daily, weekly and annual cycles; daily load has weekly and annual cycles
- SARIMA handles one integer period; MSTL decomposes several
- Holidays, Easter most of all, are the largest deviations: they need regressors
- DHR = Fourier terms + calendar dummies + ARIMA errors: flexible and transparent

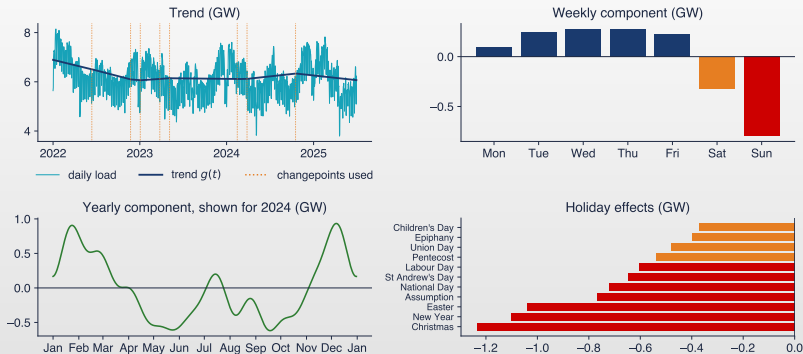
# TBATS

- De Livera, Hyndman and Snyder (2011): exponential smoothing (ETS, [Chapter 0](#)) for complex seasonality; the name lists its parts
  - ▶ trigonometric seasonality (T), Box–Cox transformation (B), ARMA errors (A), trend (T), seasonal components (S)
- $y_t^{(\omega)} = \ell_{t-1} + \phi b_{t-1} + \sum_i s_{t-1}^{(i)} + d_t$ ,  $d_t$  ARMA( $p, q$ ); level and trend updated as in Holt's method
  - ▶  $y_t^{(\omega)}$ : the series after a Box–Cox transformation with parameter  $\omega$ ;  $\ell_t$ : level;  $b_t$ : slope;  $\phi$ : damping factor;  $s_t^{(i)}$ : seasonal component  $i$
  - ▶ each period  $m_i$  has  $K_i$  harmonics  $s_{j,t}^{(i)}$ , each a rotating pair (cosine, sine) updated by the errors: a Fourier pattern that can evolve
- Automatic choices by AIC: Box–Cox or not, trend or damped trend or none, the  $K_i$ , the ARMA orders
  - ▶ daily load to June 2025, periods 7 and 365.25 (no Box–Cox, no trend):  $K = 3$  weekly and  $K = 9$  annual harmonics, ARMA(1,2), 28 states
- Limits: no regressors, hence no holidays or Easter; slow on long series; Python package `tbats`

## Prophet

- Taylor and Letham (2018) (Meta): a regression in time,  $y(t) = g(t) + s(t) + h(t) + \varepsilon_t$ 
  - ▶  $g(t)$ : piecewise-linear (or logistic) trend whose slope may change at **changepoints**; a Laplace prior shrinks most changes to zero
  - ▶  $s(t)$ : Fourier terms, by default  $K = 10$  for the year and  $K = 3$  for the week
  - ▶  $h(t)$ : one effect per holiday (with optional windows around it), from a list of dates
- Estimated by Stan (maximum a posteriori); intervals simulate future changepoints and the noise
  - ▶ no autoregressive part: today's surprise does not move tomorrow's forecast
- Built for many business series with strong calendar effects and missing data; Python package prophet

## Prophet components for daily load



□ Prophet with Romanian public holidays, daily load 2022 – June 2025

## Interpreting the Prophet components

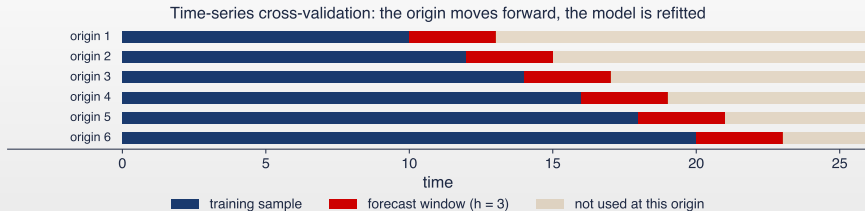
- Trend from 6.89 to 6.07 GW, flatter after 2023; 25 candidate changepoints, most shrunk to zero
  - ▶ weekly: Wednesday +0.27, Saturday -0.32, Sunday -0.80 GW
- Yearly: highest in early December (+0.93 GW, 7 December 2024), lowest in mid-September (-0.62 GW, 18 September 2024), shown for 2024
- Holidays: Christmas -1.24, New Year -1.10, Easter -1.04 GW; Children's Day only -0.37 GW
- Each component is readable and can be checked against what we know about the economy

## Recap: DHR, TBATS and Prophet side by side

|                      | DHR                           | TBATS             | Prophet          |
|----------------------|-------------------------------|-------------------|------------------|
| Seasonality          | Fourier, fixed                | Fourier, evolving | Fourier, fixed   |
| Several periods      | yes                           | yes               | yes              |
| Holidays, regressors | yes                           | no                | yes              |
| Short-run dynamics   | ARMA errors                   | ARMA errors       | none             |
| Trend                | through the errors<br>(ARIMA) | local (Holt)      | piecewise linear |
| Speed                | fast                          | slow              | fast             |

- All three are regressions on sines and cosines at heart; they differ in what else they model
- Which forecasts best is an empirical question: the next section answers it out of sample

## Time-series cross-validation



- ❑ Rolling forecast origin ([Tashman \(2000\)](#); [FPP3, Section 5.10](#)): fit on the data up to the origin, forecast  $h$  steps, move the origin, refit
- ❑ Expanding window (shown) or sliding window of fixed length; never use future data to fit

## Interpreting the design of our experiment

- Daily load: 26 origins every 14 days, from 30 June 2025 to 15 June 2026; horizon  $h = 14$  days
  - ▶ one year of out-of-sample forecasts, every season and every holiday, Easter 2026 included
- Seven methods, all refitted at each origin
  - ▶ seasonal naive (period 7); ETS (Holt–Winters, damped trend, period 7); SARIMA(2,0,1)(0,1,1)<sub>7</sub>; DHR; TBATS; Prophet
  - ▶ Combination: the simple average of the five model forecasts (all except the seasonal naive)
- The orders of SARIMA and of the DHR errors are chosen once, by AIC, on the data before the first origin: no peeking at the test period
- Cross-validation that ignores time order (random folds) is valid only for pure autoregressions with white-noise errors (Bergmeir, Hyndman and Koo (2018))

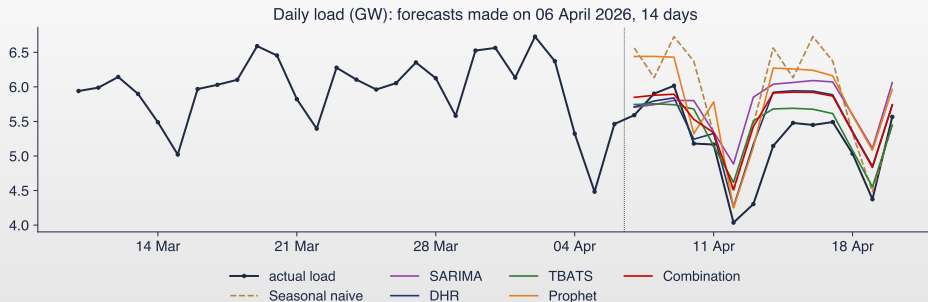
## The Diebold–Mariano test

- Diebold and Mariano (1995): is the difference in accuracy between two forecasts larger than chance?
  - ▶  $e_{1t}$ ,  $e_{2t}$ : the errors of the two forecasts;  $L$ : the loss function
  - ▶ loss differential  $d_t = L(e_{1t}) - L(e_{2t})$ ,  $L(e) = e^2$  or  $|e|$ ;  $H_0: E[d_t] = 0$
- $DM = \frac{\bar{d}}{\sqrt{\hat{V}/n}}$ ,  $\hat{V} = \hat{\gamma}_d(0) + 2 \sum_{k=1}^{h-1} \hat{\gamma}_d(k)$ 
  - ▶  $\bar{d}$ : the mean of the  $n$  differentials;  $\hat{\gamma}_d(k)$ : their autocovariance at lag  $k$ ; a negative DM favours forecast 1
  - ▶  $h$ -step errors overlap, so  $d_t$  is autocorrelated up to lag  $h - 1$
  - ▶ asymptotically  $N(0, 1)$ ; small samples:  $HLN = \sqrt{\frac{n+1-2h+h(h-1)/n}{n}}$  DM, compared with  $t(n-1)$  (Harvey, Leybourne and Newbold (1997))
- Our design: one value per origin,  $d_j$  = average loss difference over the 14-day window; the windows do not overlap, so  $h = 1$  in the formula
  - ▶ the test compares forecasts, not models: re-estimation noise is part of what is tested

## Worked example: a Diebold–Mariano test

- Five windows; window-average differences in squared error, method 1 minus method 2:  
 $-0.30, 0.10, -0.50, -0.20, -0.40$ 
  - ▶  $\bar{d} = -0.26$ : method 1 has the smaller losses in four windows out of five
- $\hat{\gamma}_d(0) = \frac{1}{5} \sum (d_j - \bar{d})^2 = 0.042$ 
  - ▶  $DM = -0.26 / \sqrt{0.042/5} = -2.82$ ;  $HLN = \sqrt{4/5} \cdot DM = -2.53$
- $|-2.53| < t_{0.975}(4) = 2.78$  ( $p = 0.065$ ): equal accuracy is not rejected at 5%
  - ▶ the normal critical value 1.96 would have rejected: with few windows, the correction matters

## Forecasts from one origin, across Easter 2026

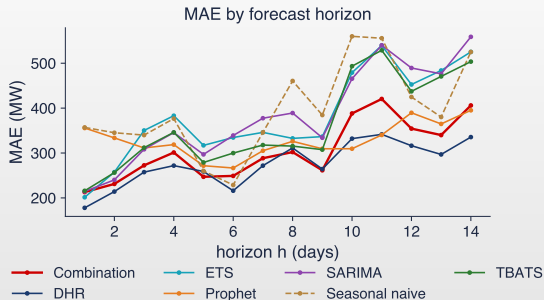
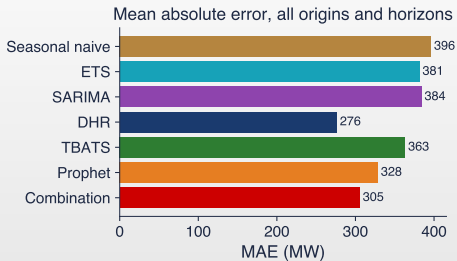


□ Origin 6 April 2026, 14 days ahead; Orthodox Easter Sunday was 12 April 2026

## Interpreting one forecast window

- MAE in this window (MW): ETS 299, TBATS 309, DHR 344, Combination 394, SARIMA 584, Prophet 616, seasonal naive 686
  - ▶ the seasonal naive method copies the previous normal week into Easter week
- Every model sees the Sunday dip; DHR and Prophet also lower Good Friday and Easter Monday, which the weekly models treat as working days
- After Easter the load stays below every forecast for the rest of the week; ETS and TBATS, whose level follows the latest days, lose least here
- One window is anecdote; the next slides average over all 26 windows

## Cross-validation results for daily load



26 origins  $\times$  14 horizons for each method; left: overall MAE; right: MAE by horizon

## Interpreting the cross-validation results

| Method         | MAE        | RMSE       | MASE        | DM against s. naive | DM against DHR |
|----------------|------------|------------|-------------|---------------------|----------------|
| Seasonal naive | 396        | 529        | 1.28        | —                   | 4.42 (< 0.001) |
| ETS            | 381        | 492        | 1.23        | −0.80 (0.431)       | 3.08 (0.005)   |
| SARIMA         | 384        | 477        | 1.24        | −1.23 (0.231)       | 3.29 (0.003)   |
| DHR            | <b>276</b> | <b>350</b> | <b>0.89</b> | −4.42 (< 0.001)     | —              |
| TBATS          | 363        | 468        | 1.18        | −1.50 (0.145)       | 3.02 (0.006)   |
| Prophet        | 328        | 412        | 1.06        | −2.99 (0.006)       | 2.76 (0.011)   |
| Combination    | 305        | 380        | 0.99        | −3.82 (< 0.001)     | 1.89 (0.070)   |

- MAE and RMSE in MW; MASE ([Hyndman and Koehler \(2006\)](#)) = MAE / 309 MW, the in-sample MAE of the weekly seasonal naive method; DM: HLN statistic (p-value) on window-average squared errors, row method minus column method
- DHR wins; with the combination, it is the only method with MASE below 1; Prophet is the second single model: the two that know the holidays
- ETS, SARIMA and TBATS are not significantly better than the seasonal naive method

## A second test: Romanian GDP

| Method                    | MAE, $h = 1$ | MAE, $h = 4$ | without 2019–2020, $h = 1$ | without 2019–2020, $h = 4$ |
|---------------------------|--------------|--------------|----------------------------|----------------------------|
| SARIMA (airline)          | 1.69         | 2.70         | 1.32                       | 1.89                       |
| ETS                       | 1.76         | 2.84         | 1.41                       | 2.44                       |
| Seasonal naive            | 3.81         | 3.74         | 3.80                       | 3.49                       |
| Combination (SARIMA, ETS) | 1.70         | 2.65         | 1.34                       | 2.02                       |

- 100 ln Y, errors in %; 51 quarterly origins, 2012Q4–2025Q2, expanding window; 43 origins without 2019–2020
- DM (HLN), SARIMA against ETS:  $h = 1$ :  $-1.46$  ( $p = 0.150$ );  $h = 4$ :  $0.50$  ( $p = 0.620$ )
- SARIMA against the seasonal naive method:  $h = 1$ :  $-4.00$  ( $p < 0.001$ );  $h = 4$ :  $-0.54$  ( $p = 0.591$ )
- One year ahead, the pandemic errors dominate and no difference is significant; without them, SARIMA has the smallest MAE at  $h = 4$

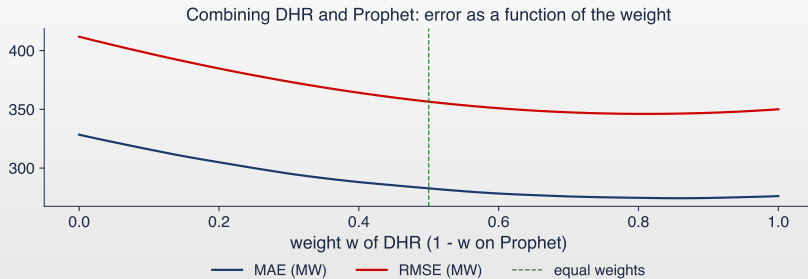
## Recap: Forecast evaluation

- Evaluate out of sample with rolling origins; choose model orders before the test period
- MASE compares with the seasonal naive method and across series
- The DM test (with the HLN correction) says whether a gain is more than chance
- Daily load: the models that know the calendar (DHR, Prophet) win; for GDP, SARIMA and ETS are close

## Why combining forecasts works

- **Bates and Granger (1969)**: a weighted average of two unbiased forecasts,  $f_c = wf_1 + (1 - w)f_2$ 
  - ▶  $w$ : the weight of forecast 1;  $\sigma_1, \sigma_2$ : the standard deviations of the two errors;  $\rho$ : their correlation
  - ▶  $MSE(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\rho\sigma_1\sigma_2$
  - ▶ best weight  $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$ ; the gain is large when  $\rho$  is small
- **Example**:  $\sigma_1 = 1, \sigma_2 = 1.2, \rho = 0.75$ :  $MSE(0.5) = 1.060$ ;  $w^* = 0.84$ ,  $MSE(w^*) = 0.984$ 
  - ▶ highly correlated errors: the equal-weight average is worse than the better forecast ( $\sigma_1^2 = 1$ ), and even the best weight gains little
- **The forecast combination puzzle** (**Stock and Watson (2004)**; **Smith and Wallis (2009)**): estimated weights rarely beat equal weights
  - ▶ the weights must be estimated, and their sampling error eats the theoretical gain

## Combining DHR and Prophet



□ Cross-validation errors of the daily load;  $w$  is the weight of DHR

## Interpreting the combination

- MAE: Prophet alone ( $w = 0$ ) 328 MW, DHR alone ( $w = 1$ ) 276 MW, equal weights 283 MW
  - ▶ best in hindsight:  $w = 0.86$ , MAE 274 MW, a small gain chosen after seeing the test data
- The errors of the two methods are strongly correlated ( $\rho = 0.75$ ): both miss the same cold spells
- The five-model average (MAE 305) is second only to DHR and better than four of its five members: cheap insurance when the winner is not known in advance
- In the M4 competition ([Makridakis, Spiliotis and Assimakopoulos \(2020\)](#)), most of the best methods were combinations; review: [Wang et al. \(2023\)](#)

## Possible contribution of AI

- ▣ **Code:** a first draft of a rolling-origin loop that refits SARIMA, DHR, TBATS and Prophet and stores the errors by horizon
- ▣ **Calendars:** lists of holidays and moving feasts for other countries, to be checked against official sources
- ▣ **Explanation:** a second derivation of the ACF of the airline model, or of the eventual forecast function
- ▣ Example prompt
  - ▶ Write Python code that forecasts daily electricity load 14 days ahead with a regression on annual Fourier terms, weekday dummies and Romanian public holidays (Orthodox Easter from dateutil) with ARMA errors, and evaluates it against SARIMA by rolling origins and a Diebold-Mariano test.

## Checks you must run

- ▣ The Easter date: Orthodox, not Western (EASTER\_ORTHODOX); in 2024 the two differ by five weeks
- ▣ No look-ahead: features, model orders and holiday lists must be available at each origin
- ▣ The sign conventions of  $\theta$  and  $\Theta$  in the software, and the degrees of freedom of Ljung–Box
- ▣ Adjusted or unadjusted data: SARIMA on NSA; growth rates on the right version
- ▣ That the DM test uses losses from the same windows for both methods, and the HLN correction for small  $n$
- ▣ Every cited reference: it must exist; check the DOI

## Key takeaways

- ▣ Seasonality repeats with a known period; it can be deterministic (dummies, Fourier) or stochastic ( $\Delta_s$ )
- ▣ SARIMA multiplies regular and seasonal polynomials; the airline model fits many series with two parameters
- ▣ Official data: SCA for short-run growth, NSA with calendar regressors for SARIMA forecasting
- ▣ Calendar effects (working days, Orthodox Easter, holidays) are known in advance and often matter more than the model class
- ▣ Multiple seasonality: MSTL, DHR, TBATS, Prophet; on Romanian daily load, DHR wins out of sample
- ▣ Judge forecasts by rolling origins, MASE and the DM test; combine them when the winner is uncertain

## Key formulas

| Quantity            | Formula                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Seasonal difference | $\Delta_s y_t = (1 - L^s)y_t, \quad \Delta \Delta_s = 1 - L - L^s + L^{s+1}$                                        |
| SARIMA              | $\phi(L)\Phi(L^s)(1 - L)^d(1 - L^s)^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$                                      |
| Airline             | $\rho_1 = \frac{\theta}{1+\theta^2}, \quad \rho_s = \frac{\Theta}{1+\Theta^2}, \quad \rho_{s\pm 1} = \rho_1 \rho_s$ |
| Fourier             | $\sum_{k=1}^K [\alpha_k \sin(2\pi kt/m) + \beta_k \cos(2\pi kt/m)]$                                                 |
| Prophet             | $y(t) = g(t) + s(t) + h(t) + \varepsilon_t$                                                                         |
| MASE                | $\frac{1}{h} \sum  e_{T+j}  / \frac{1}{T-m} \sum  y_t - y_{t-m} $                                                   |
| DM                  | $\bar{d} / \sqrt{\hat{V}/n}, \quad d_t = L(e_{1t}) - L(e_{2t})$                                                     |
| Combination         | $w^* = \frac{\sigma_2^2 - \rho \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2}$               |

## Self-assessment

- **Question:** how many parameters does  $\text{SARIMA}(1, 1, 1)(0, 1, 1)_{12}$  have, and at which lags does  $\varepsilon$  appear?
  - ▶ **Answer:**  $\phi, \theta, \Theta, \sigma^2$ , four;  $\varepsilon$  at lags 0, 1, 12, 13 (the last with coefficient  $\theta\Theta$ )
- **Question:** unadjusted GDP falls by 40% from Q4 to Q1. Is this a recession?
  - ▶ **Answer:** no; it is the seasonal pattern; look at the SCA series or at year-on-year growth
- **Question:** a DM test with 26 windows gives  $\text{HLN} = -1.93$ . Is the first forecast significantly better at 5%?
  - ▶ **Answer:** no:  $|-1.93| < t_{0.975}(25) = 2.06$ ; it is better on average, but not significantly
- **Project idea:** forecast Romanian hourly electricity load with a DHR (daily, weekly and annual Fourier terms, holidays, temperature) against TBATS, Prophet and a combination, by rolling origins and DM tests
  - ▶ Next: [Chapter 5](#), conditional volatility: ARCH and GARCH models

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